

WASP-36b

R-1089

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_170205 · created 2026-07-30T16:56:47+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457789.7899 +/- 0.0022 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.163 +/- 0.048
Transit depth (Rp/Rs)^2	2.66 +/- 1.56 [%]
Orbital Inclination (inc)	83.07 +/- 2.82
Airmass coefficient 1 (a1)	0.91 +/- 0.51
Airmass coefficient 2 (a2)	0.11 +/- 0.8
Scatter in the residuals of the lightcurve fit is	60.57 %
Best Comparison Star	None
Optimal Aperture	2.53
Optimal Annulus	6.0
Transit Duration (day)	0.072 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.163 ± 0.048 vs 0.1368 (deviation 0.53σ)
Duration fit vs published	0.072 d vs 0.0758 d (deviation 0.15σ)
Mid-time O–C	-5.7 min
Depth SNR	3.3
Residual scatter	60.57 %

Light curve

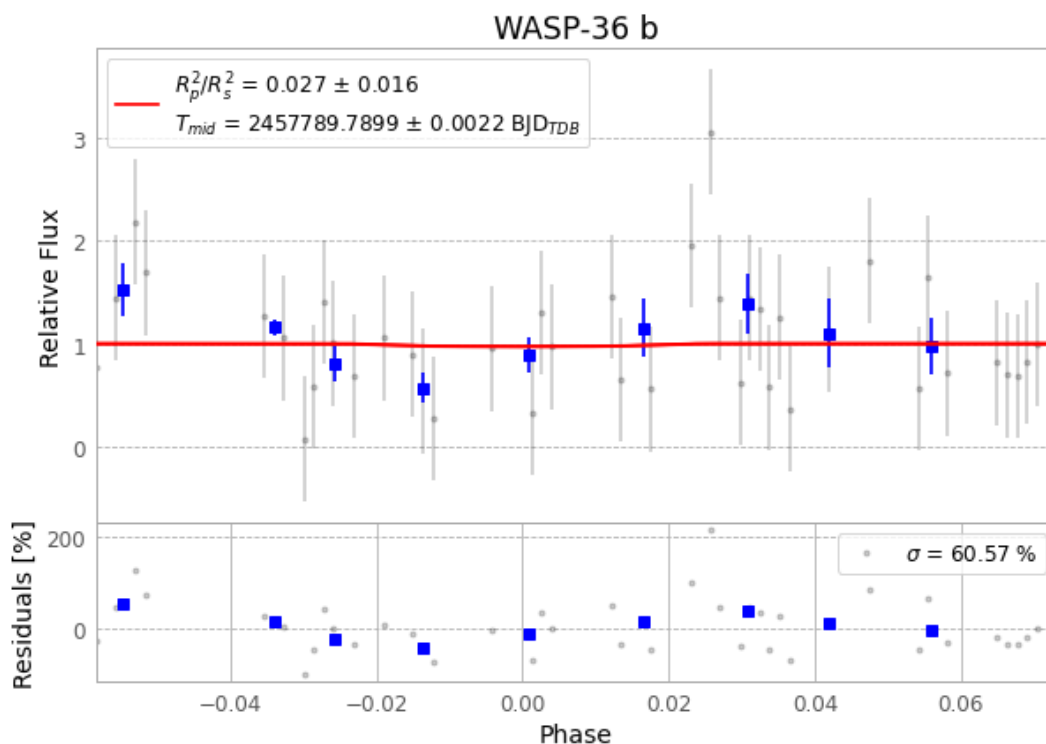


Figure 1 — FinalLightCurve_WASP-36 b_2017-02-04.png (SHA-256 21b63b587a910afe...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [418, 412], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 76694a1588f60680...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

96 FITS frames (63 MB), WASP-36170205043912.FITS → WASP-36170205093012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-170205.log (c673d49ca018...), FinalLightCurve_WASP-36 b_2017-02-04.png (21b63b587a91...), FinalLightCurve_WASP-36 b_2017-02-04.csv (7f73b4cc2d9c...)

Compute resources

Wall time 165.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 16:56Z.